PROPOSAL REPORT

RIDELANKA FLEETHIRE SYSTEM

Web based Car Rental Management System

MODULE	GROUP	SUBMISSION DATE
SE2030 SOFTWARE ENGINEERING	2026-Y2-S1-KU-36 SIX MEMBERS - SIX CORE FUNCTIONS	07 - AUGUST - 2026

GROUP MEMBERS AND CORE OWNERSHIP

01 S M H C B D Rambukpitiya - IT25103736
User & Customer Management

02 I K S P Jayawardana - IT25103732
Vehicle Management

03 W M K K G N A Weerasingha - IT25103737
Booking & Reservation Management

04 R M M D Bhagyamathi - IT25103733
Payment Management

05 R M R D Rajapaksha - IT25103734
Feedback & Support Management

06 K A K Dananjaya - IT25103735
Fleet Management

1. Introduction

1.1 Project Overview

RideLanka FleetHire is a comprehensive web-based car rental management system designed to transform fragmented manual rental operations into a connected, efficient digital platform. The system addresses the critical challenges faced by car rental businesses operating across multiple branches, where disconnected systems, manual booking registers, spreadsheet-based fleet tracking, and phone-based reservations create operational inefficiencies and poor customer experiences.

The platform serves as a single trusted source for managing vehicles, bookings, customers, payments, feedback, and maintenance activities. By centralizing these operations, RideLanka FleetHire enables real-time vehicle availability tracking, automated booking confirmations, streamlined payment processing, and comprehensive customer journey management.

1.2 Project Objectives

The RideLanka FleetHire project aims to achieve four primary objectives:

Centralize Operations: Connect vehicles, bookings, customers, payments, feedback, and maintenance through a unified platform, eliminating data silos and manual record-keeping.

Simplify User Experience: Make vehicle search, booking, payment, and rental access intuitive and straightforward for customers while providing staff with efficient operational tools.

Enable Rapid Response: Handle customer issues, maintenance needs, and operational changes quickly through automated notifications and streamlined workflows.

Strengthen Control: Improve financial reconciliation, reporting capabilities, permission management, and audit trails to support better business decision-making.

1.3 Target Users and Stakeholders

RideLanka FleetHire serves a diverse range of users across the rental ecosystem:

Customers: Individuals and businesses seeking vehicle rentals who require easy search, booking, payment, and rental management capabilities.

Operations Management: General managers and operations staff who need visibility and control over business operations, user accounts, and access permissions.

Fleet Management: Staff responsible for keeping vehicles organized, available, and correctly categorized across all branches.

Reservations Staff: Marketing and reservations executives managing booking pipelines and promotional campaigns.

Finance Personnel: Officers handling billing, collections, and financial reporting for rental transactions.

Customer Service: Staff managing post-rental customer experiences, including complaint handling and satisfaction tracking.

Maintenance Coordinators: Personnel responsible for keeping vehicles roadworthy through servicing and repairs.

1.4 Scope and Limitations

In Scope:

- Online vehicle search, booking, and reservation management
- Customer registration, authentication, and profile management
- Payment processing with digital receipts and confirmations
- Vehicle fleet management including availability tracking and status updates
- Booking modification, cancellation, and refund handling
- Customer feedback, rating, and complaint management
- Maintenance scheduling and service history tracking
- Role-based access control and audit trail
- Branch dashboard and operational reporting

Out of Scope (Future Enhancements):

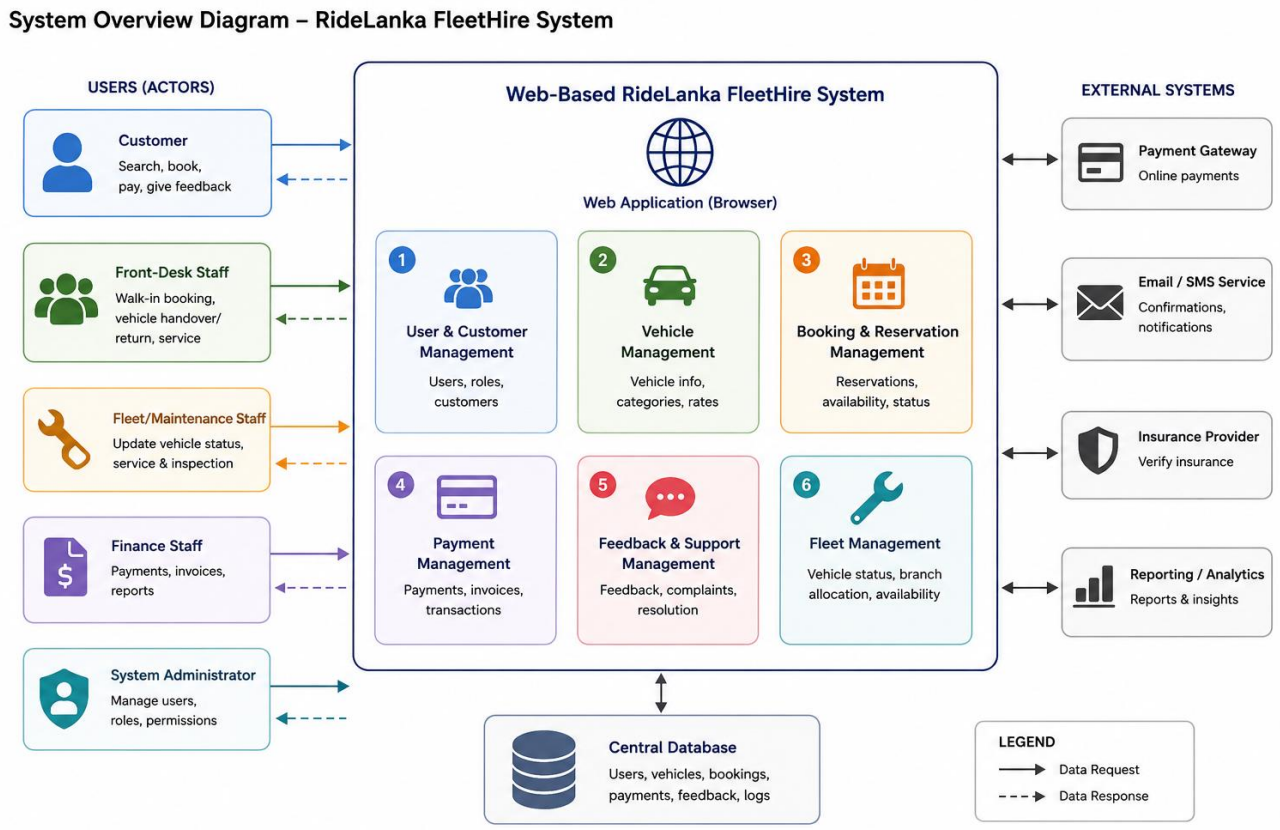
- Mobile application development (browser-based only)
- Integration with third-party travel agencies
- Loyalty and rewards program
- Advanced analytics and business intelligence
- Offline booking handling (open for future consideration)
- Multi-language support beyond initial implementation

Technical Constraints:

- Web-based platform only
- Java-based application development
- External payment gateway dependency
- SMS/email notification service dependency

2. System Overview Diagram

The RideLanka FleetHire system operates through a connected platform that links all operational modules, role-based access controls, and exception handling mechanisms.



3. Functional Requirements

The system is grouped into three functional areas: Customer Services, Staff Operations, and Exception Recovery.

Customer Services

- **Vehicle Search – By category, branch location, and availability dates.**
- **Online Booking – Create, modify, and cancel reservations via web.**
- **Vehicle Selection – Choose vehicle with add-ons and optional extras.**
- **Payment Processing – Secure payment with digital receipt and confirmation.**
- **Booking History – View past/upcoming bookings; request modifications.**
- **Promotions – Apply discount codes and promotional offers.**
- **Guest Checkout – Book without registration (proposed).**

Staff Operations

- **Fleet Control – Register, update, and deactivate vehicles.**
- **Availability Tracking – Real-time status updates and branch transfers.**
- **Booking Verification – Confirm, reconcile, and approve reservations.**
- **Refund & Incident Handling – Process refunds, complaints, and incidents.**
- **Branch Dashboard – View operational metrics and customer notices.**
- **Reports & Audit – Role-based reports, search, and activity logs.**
- **Maintenance Scheduling – Plan and track vehicle servicing and repairs.**

Exception & Recovery

- **Payment Recovery – Handle failed, pending, and duplicate transactions.**
- **Booking Recovery – Recover paid bookings that failed to confirm.**
- **Maintenance Impact – Notify affected bookings; reassign, upgrade, or refund.**
- **Notification Retry – Retry failed SMS/email alerts with staff fallback.**

4. Non-Functional Requirements

The system must meet the following quality attributes across performance, security, reliability, usability, and scalability.

Performance

- **Vehicle search response time: ≤ 3 seconds**
- **Booking confirmation after payment: ≤ 10 seconds**
- **System uptime: 99.5% target**
- **Recovery Time Objective (RTO): ≤ 4 hours**
- **Maximum data loss window: ≤ 15 minutes**

Security & Privacy

- **Credentials protected with secure session management**
- **Role-based access control with least-privilege principle**
- **Customers can view only their own data**
- **Complete audit trail for all critical actions**
- **No raw card credentials stored in the system**
- **External payment gateway dependency for payment processing**

Reliability & Recovery

- **All bookings recoverable from system failures**
- **Automated payment retry mechanisms for failed transactions**
- **Full transaction logging for audit and reconciliation**
- **Regular data backups with tested restoration procedures**

Usability & Accessibility

- **Mobile-first responsive design for all screen sizes**
- **Large, readable text with keyboard navigation support**
- **Screen-reader compatible for visually impaired users**
- **Clear, consistent user interface with intuitive navigation flow**

Scalability

- **Supports growing user base and peak load periods**
- **Compatible with major web browsers and devices**
- **Handles concurrent operations across multiple branches**

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5. Major Stakeholders and Users

Operations Management - Nuwan Perera - General Manager

- **Role:** Oversees overall business operations
- **Needs:** Full visibility and control over user accounts and access
- **System Interaction:** Dashboard views, user management, operational oversight

Fleet Management - Kasun Fernando - Fleet Operations Manager

- **Role:** Maintains fleet organization and availability
- **Needs:** Vehicle categorization, availability management, branch coordination
- **System Interaction:** Vehicle CRUD, availability updates, branch transfers

Reservations - Tharushi Madushani - Marketing & Reservations Executive

- **Role:** Manages reservation pipeline and promotions
- **Needs:** Booking management, campaign tools, promotional features
- **System Interaction:** Booking monitoring, discount management, promotion tracking

Finance - Shanika Wijeratne - Finance & Accounts Officer

- **Role:** Handles billing, collections, and financial reporting
- **Needs:** Payment verification, reconciliation, financial reporting
- **System Interaction:** Payment processing, financial reports, audit access

Customer Service - Dilini Jayasinghe - Customer Service Supervisor

- **Role:** Manages post-rental customer experience

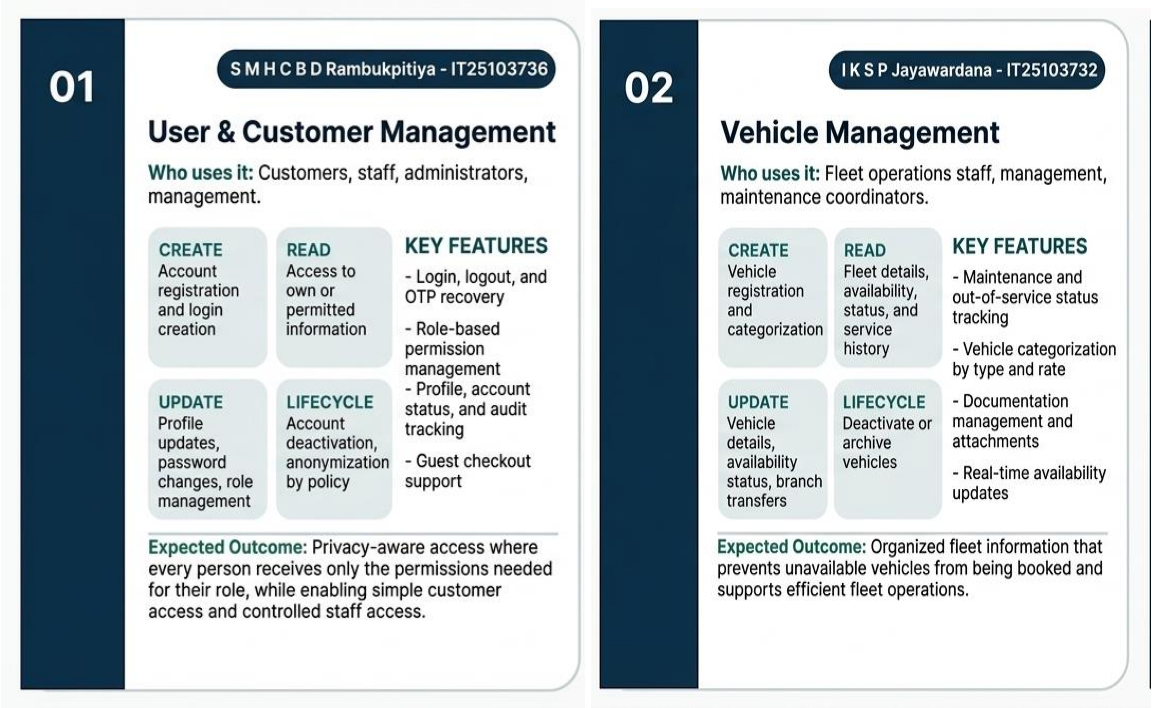
- **Needs:** Complaint handling, satisfaction tracking, issue resolution
- **System Interaction:** Feedback management, complaint resolution, satisfaction reporting

Maintenance - Ravindu Senanayake - Vehicle Maintenance Coordinator

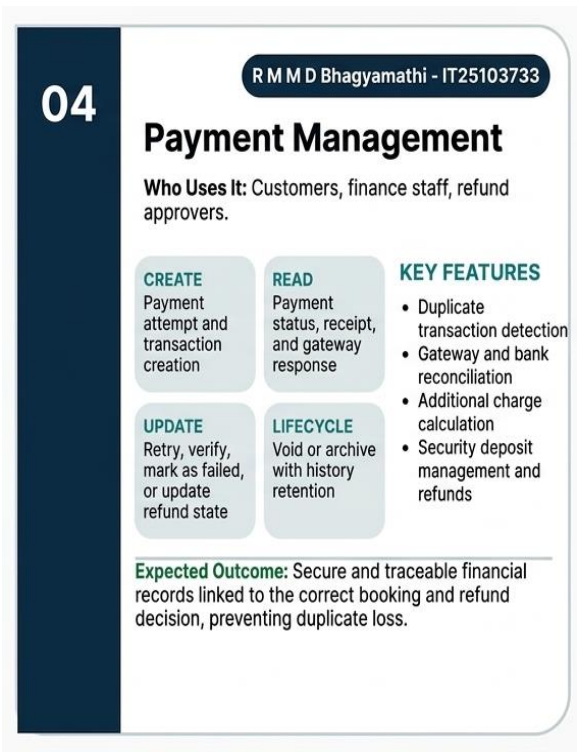
- **Role:** Coordinates vehicle servicing and repairs
- **Needs:** Maintenance scheduling, repair logging, vehicle status updates
- **System Interaction:** Maintenance planning, service history, status updates

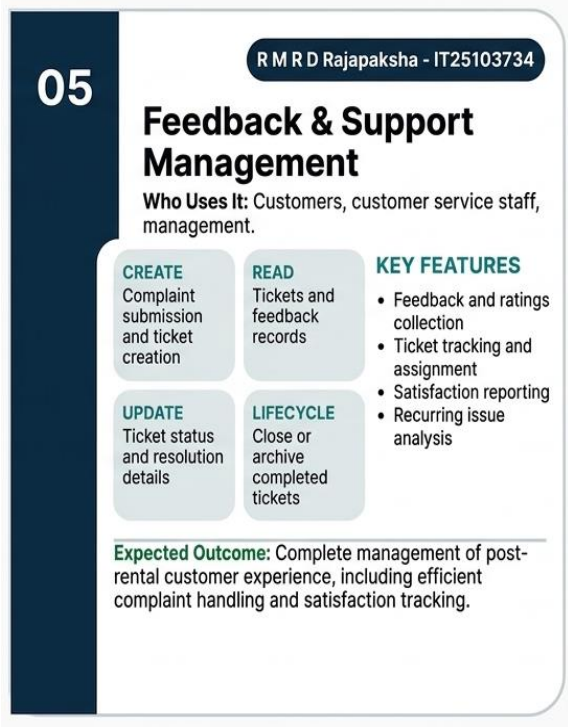
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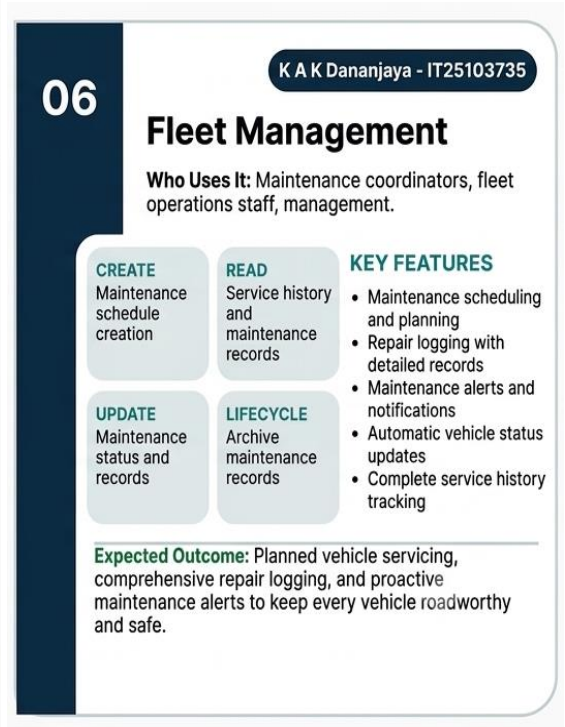
6. Six Major Functions




03
W M K K G N A Weerasingha - IT25103737
Booking & Reservation
Who Uses it: Customers, reservations staff, customer service.
CREATE
Online booking creation
READ
Booking details, status, and history
UPDATE
Amend or cancel existing reservations
LIFECYCLE
Cancel or archive with audit trail
KEY FEATURES
- Real-time availability checks
- Promotions and discount rule management
- Reservation calendar view
- Policy rule enforcement
Expected Outcome: End-to-end reservation management with real-time availability verification, promotional campaign support, and complete audit history.


04
R M M D Bhagyamathi - IT25103733
Payment Management
Who Uses It: Customers, finance staff, refund approvers.
CREATE
Payment attempt and transaction creation
READ
Payment status, receipt, and gateway response
UPDATE
Retry, verify, mark as failed, or update refund state
LIFECYCLE
Void or archive with history retention
KEY FEATURES
• Duplicate transaction detection
• Gateway and bank reconciliation
• Additional charge calculation
• Security deposit management and refunds
Expected Outcome: Secure and traceable financial records linked to the correct booking and refund decision, preventing duplicate loss.


05
R M R D Rajapaksha - IT25103734
Feedback & Support Management
Who Uses It: Customers, customer service staff, management.
CREATE
Complaint submission and ticket creation
READ
Tickets and feedback records
UPDATE
Ticket status and resolution details
LIFECYCLE
Close or archive completed tickets
KEY FEATURES
• Feedback and ratings collection
• Ticket tracking and assignment
• Satisfaction reporting
• Recurring issue analysis
Expected Outcome: Complete management of post-rental customer experience, including efficient complaint handling and satisfaction tracking.


06
K A K Dananjaya - IT25103735
Fleet Management
Who Uses It: Maintenance coordinators, fleet operations staff, management.
CREATE
Maintenance schedule creation
READ
Service history and maintenance records
UPDATE
Maintenance status and records
LIFECYCLE
Archive maintenance records
KEY FEATURES
• Maintenance scheduling and planning
• Repair logging with detailed records
• Maintenance alerts and notifications
• Automatic vehicle status updates
• Complete service history tracking
Expected Outcome: Planned vehicle servicing, comprehensive repair logging, and proactive maintenance alerts to keep every vehicle roadworthy and safe.

7. Minor Functions

These supporting features strengthen the platform across three areas and are not extra team-member allocations.

Customer Service Support

- Online Booking – Web-based reservations
- Vehicle Search – By category, location, dates
- Booking Modifications – Change or cancel bookings
- Feedback Submission – Ratings and reviews
- Digital Receipts – E-payment confirmation
- Booking History – View past and upcoming rentals

Operations Support

- Branch Dashboard – Real-time operational view
- Vehicle Status – Availability and condition updates
- Customer Communications – Manage notifications
- Availability Updates – Fleet status changes
- Branch Transfers – Move vehicles between branches
- Walk-in Handling – Process in-person rentals

Control & Data Support

- Role-Based Access – Permission-controlled system access
- Activity Logs – Track all system actions
- Reporting & Analytics – Business performance reports
- Audit Trail – Complete change history
- Data Backup – Scheduled system backups
- Policy Enforcement – Apply business rules

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8. System Limitations and Constraints

Technical Boundaries

- **Platform:** Web-based application only; no mobile app.
- **Payment Gateway:** External gateway dependency.
- **Notifications:** SMS and email service dependency.

- **Browser Support:** Modern web browsers only.
- **Device Support:** Responsive design for desktop and mobile browsers.
- **Integration:** Limited to specified external services.

Operational Constraints

- **Access Control:** Role-limited access with audit trail.
- **Payment Verification:** Verified payment required before booking confirmation.
- **Credit Card Security:** No raw card credentials stored.
- **Refund Authority:** Refunds require finance authorization.
- **Reporting:** Real company financial reports remain confidential.

Decisions Required

Security Deposit Handling · Refund Thresholds & Limits · Late Return Fee Calculation · Damage Charge Assessment · Customer Verification · Admin & Incident Permissions · Peak Users & Devices · Vendor Selection · Deletion Rules & Retention *(all open for decision).*

Assumptions

- **Internet Access:** Users have reliable internet access.
- **Device Compatibility:** Users access via modern web browsers.
- **Payment Gateway:** Gateway is available and integrated.
- **Staff Training:** Staff receive training on system usage.
- **Network Security:** Secure network infrastructure in place.

9. Project Timeline

Proposed Week 1–14 Delivery Plan

Week	Phase	Planned Activity	Expected Output
01	Initiation	Team formation, project kickoff, ownership assignment	Team structure and roles
02	Discovery	Requirements discovery and stakeholder interviews	Initial requirements
03	Discovery	Requirements validation and baseline confirmation	Proposal delivery
04	Design	Design baseline and architecture definition	Design approach
05	Design	Requirements specification and business process modeling	SRS document
06	Design	Architecture, ERD, UI design, and test planning	Design documentation
07	Build	Core build - Modules 1-2 (User & Vehicle)	Working module progress
08	Build	Core feature completion - Module 1-2	Complete modules 1-2
09	Build	Core build - Modules 3-4 (Booking & Payment)	Working module progress

Week	Phase	Planned Activity	Expected Output
10	Build	Core build - Modules 5-6 (Feedback & Fleet)	Progress presentation
11	Integration	Integration of all modules, reporting	Integrated system
12	Test	Security, accessibility, and performance testing	Test results
13	Verification	UAT, defect fixing, documentation	Submission candidate
14	Submission	Final demo, report submission, and team presentation	Final submission

Visual Delivery Map

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Week 03 - Proposal → Requirements baseline validated

Week 07 - Core Build → Modules 1-2 development complete

Week 10 - Progress → 75% system functionality complete

Week 11 - Integration → Six modules connected through shared database

Week 13 - UAT → Complete system testing and fixes

Week 14 - Submission → Final system, report, and presentation

Weekly Rhythm: Each owner reports completed work, blockers, shared-data changes, and next-week tasks so module progress stays coordinated.

10. Conclusion

Why RideLanka FleetHire Is Worth Implementing

RideLanka FleetHire addresses the critical challenges faced by car rental businesses operating across multiple branches with disconnected manual systems. The platform transforms fragmented operations into a connected, efficient digital ecosystem that benefits all stakeholders.

Practical Value:

The system connects fragmented rental operations through one trusted platform, eliminating manual booking registers, spreadsheet-based fleet tracking, and phone-based reservations. This centralization enables:

- **Simple Rentals:** Clear search, booking, payment, digital receipts, and customer updates
- **Connected Operations:** Shared vehicles, availability, branch transfers, maintenance, and customer information
- **Trusted Decisions:** Role access, reconciliation, reports, recovery, and audit history

Feasibility:

The project is structured as six core functions assigned to individual team members, with supporting features shared across the platform. This modular approach enables:

- Clear ownership and accountability
- Parallel development and integration
- Phased delivery with visible progress
- Practical application of software engineering principles

Completion Approach:

The team will follow a structured development process:

1. **Discover:** Validate requirements and complete the proposal
2. **Design:** Create architecture, ERD, UI design, and test plans
3. **Build:** Develop core functions with weekly coordination
4. **Verify:** Test, fix defects, and prepare for submission
5. **Deliver:** Present the completed system and final documentation

Next Steps:

- Approve remaining policy, capacity, and integration decisions
- Freeze the design and begin implementation
- Maintain weekly coordination and progress reporting

One Platform • Six Core Modules • One Reliable Ride